

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

Let $\{x\}$ denote the fractional part of x and $f(x) = \frac{\cos^{-1}(1-\{x\}^2) \sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}$, $x \neq 0$. If L and R respectively denotes the left hand limit and the right hand limit of $f(x)$ at $x = 0$, then $\frac{32}{\pi^2}(L^2 + R^2)$ is equal to

Q2 - 2024 (01 Feb Shift 2)

Let $f(x) = \begin{cases} x - 1, & x \text{ is even,} \\ 2x, & x \text{ is odd,} \end{cases} x \in \mathbb{N}$. If for some $a \in \mathbb{N}$, $f(f(f(a))) = 21$, then $\lim_{x \rightarrow a^-} \left\{ \frac{|x|^3}{a} - \left[\frac{x}{a} \right] \right\}$,

where $[t]$ denotes the greatest integer less than or equal to t , is equal to :

(1) 121

(2) 144

(3) 169

(4) 225

Q3 - 2024 (27 Jan Shift 1)

If $a = \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}}-\sqrt{2}}{x^4}$ and $b = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2}-\sqrt{1+\cos x}}$, then the value of ab^3 is :

(1) 36

(2) 32

(3) 25

(4) 30

Q4 - 2024 (27 Jan Shift 2)

If $\lim_{x \rightarrow 0} \frac{3+\alpha \sin x + \beta \cos x + \log_e(1-x)}{3 \tan^2 x} = \frac{1}{3}$, then $2\alpha - \beta$ is equal to :

(1) 2

(2) 7

(3) 5

(4) 1

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Q5 - 2024 (29 Jan Shift 1)

$\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1}{\left(x - \frac{\pi}{2}\right)^2} \int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \cos\left(\frac{1}{t^3}\right) dt \right)$ is equal to

(1) $\frac{3\pi}{8}$

(2) $\frac{3\pi^2}{4}$

(3) $\frac{3\pi^2}{8}$

(4) $\frac{3\pi}{4}$

Q6 - 2024 (29 Jan Shift 2)

Let the slope of the line $45x + 5y + 3 = 0$ be $27r_1 + \frac{9r_2}{2}$ for some $r_1, r_2 \in R$. Then

$\lim_{x \rightarrow 3} \left(\int_3^x \frac{8t^2}{\frac{3r_2x}{2} - r_2x^2 - r_1x^3 - 3x} dt \right)$ is equal to _____

Q7 - 2024 (30 Jan Shift 1)

Let $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow R$ be a differentiable function such that $f(0) = \frac{1}{2}$. If the $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{e^{x^2} - 1} = \alpha$, then $8\alpha^2$ is equal to :

(1) 16

(2) 2

(3) 1

(4) 4

Q8 - 2024 (31 Jan Shift 1)

Let a be the sum of all coefficients in the expansion of $(1 - 2x + 2x^2)^{2023} (3 - 4x^2 + 2x^3)^{2024}$ and

$b = \lim_{x \rightarrow 0} \left(\frac{\int_0^x \frac{\log(1+t)}{t^{2024} + 1} dt}{x^2} \right)$. If the equations $cx^2 + dx + e = 0$ and $2bx^2 + ax + 4 = 0$ have a common root,

where $c, d, e \in R$, then $d : c : e$ equals

(1) 2 : 1 : 4

(2) 4 : 1 : 4

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(3) 1 : 2 : 4

(4) 1 : 1 : 4

Q9 - 2024 (31 Jan Shift 1)

$$\lim_{x \rightarrow 0} \frac{e^{4 \sin x} - 2|\sin x| - 1}{x^2}$$

(1) is equal to -1

(2) does not exist

(3) is equal to 1

(4) is equal to 2

Q10 - 2024 (31 Jan Shift 2)

Let $f : \mathbb{R} \rightarrow (0, \infty)$ be strictly increasing function such that $\lim_{x \rightarrow \infty} \frac{f(7x)}{f(x)} = 1$. Then, the value of

$$\lim_{x \rightarrow \infty} \left[\frac{f(5x)}{f(x)} - 1 \right]$$
 is equal to

(1) 4

(2) 0

(3) 7/5

(4) 1

Q11 - 2024 (31 Jan Shift 2)

$$\lim_{x \rightarrow 0} \frac{ax^2e^x - b \log_e(1+x) + cxe^{-x}}{x^2 \sin x} = 1 \text{ then } 16(a^2 + b^2 + c^2) \text{ is equal to } \underline{\hspace{2cm}}.$$

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Answer Key					
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Q1 (18)	//. mathongo	Q2 (2)	//. mathongo	Q3 (2)	//. mathongo
Q4 (3)	//. mathongo	Q5 (3)	//. mathongo	Q6 (12)	//. mathongo
Q7 (2)	//. mathongo	Q8 (4)	//. mathongo	Q9 (4)	//. mathongo
Q10 (2)	//. mathongo	Q11 (81)	//. mathongo	//. mathongo	//. mathongo
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Solutions

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Q1

Finding right hand limit

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h)$$

$$= \lim_{h \rightarrow 0} f(h)$$

$$= \lim_{h \rightarrow 0} \frac{\cos^{-1}(1 - h^2) \sin^{-1}(1 - h)}{h(1 - h^2)}$$

$$= \lim_{h \rightarrow 0} \frac{\cos^{-1}(1 - h^2)}{h} \left(\frac{\sin^{-1} 1}{1} \right)$$

$$\text{Let } \cos^{-1}(1 - h^2) = \theta \Rightarrow \cos \theta = 1 - h^2$$

$$= \frac{\pi}{2} \lim_{\theta \rightarrow 0} \frac{\theta}{\sqrt{1 - \cos \theta}}$$

$$= \frac{\pi}{2} \lim_{\theta \rightarrow 0} \frac{1}{\sqrt{\frac{1 - \cos \theta}{\theta^2}}}$$

$$= \frac{\pi}{2} \frac{1}{\sqrt{1/2}}$$

$$R = \frac{\pi}{\sqrt{2}}$$

Now finding left hand limit

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Solutions

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$$\begin{aligned}
 L &= \lim_{x \rightarrow 0^-} f(x) \\
 &= \lim_{h \rightarrow 0} f(-h) \\
 &= \lim_{h \rightarrow 0} \frac{\cos^{-1}(1 - \{-h\}^2) \sin^{-1}(1 - \{-h\})}{\{-h\} - \{-h\}^3} \\
 &= \lim_{h \rightarrow 0} \frac{\cos^{-1}(1 - (-h + 1)^2) \sin^{-1}(1 - (-h + 1))}{(-h + 1) - (-h + 1)^3} \\
 &= \lim_{h \rightarrow 0} \frac{\cos^{-1}(-h^2 + 2h) \sin^{-1} h}{(1 - h)(1 - (1 - h)^2)} \\
 &= \lim_{h \rightarrow 0} \left(\frac{\pi}{2} \right) \frac{\sin^{-1} h}{(1 - (1 - h)^2)} \\
 &= \frac{\pi}{2} \lim_{h \rightarrow 0} \left(\frac{\sin^{-1} h}{-h^2 + 2h} \right) \\
 &= \frac{\pi}{2} \lim_{h \rightarrow 0} \left(\frac{\sin^{-1} h}{h} \right) \left(\frac{1}{-h + 2} \right) \\
 L &= \frac{\pi}{4} \\
 \frac{32}{\pi^2} (L^2 + R^2) &= \frac{32}{\pi^2} \left(\frac{\pi^2}{2} + \frac{\pi^2}{16} \right) \\
 &= 16 + 2 \\
 &= 18
 \end{aligned}$$

Q2

$$\begin{aligned}
 f(x) &= \begin{cases} x - 1; & x = \text{even} \\ 2x; & x = \text{odd} \end{cases} \\
 f(f(f(a))) &= 21 \\
 \text{C-1: If } a &= \text{even} \\
 f(a) &= a - 1 = \text{odd} \\
 f(f(a)) &= 2(a - 1) = \text{even} \\
 f(f(f(a))) &= 2a - 3 = 21 \Rightarrow a = 12 \\
 \text{C-2: If } a &= \text{odd} \\
 f(a) &= 2a = \text{even} \\
 f(f(a)) &= 2a - 1 = \text{odd} \\
 f(f(f(a))) &= 4a - 2 = 21 \text{ (Not possible)} \\
 \text{Hence } a &= 12
 \end{aligned}$$

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Solutions

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Now

$$\begin{aligned} \lim_{x \rightarrow 12^-} \left(\frac{|x|^3}{2} - \left[\frac{x}{12} \right] \right) \\ = \lim_{x \rightarrow 12^-} \frac{|x|^3}{2} - \lim_{x \rightarrow 12^-} \left[\frac{x}{12} \right] \\ = 144 - 0 = 144. \end{aligned}$$

Q3

$$\begin{aligned} a &= \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}} - \sqrt{2}}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{\sqrt{1+x^4} - 1}{x^4 \left(\sqrt{1+\sqrt{1+x^4}} + \sqrt{2} \right)} \\ &= \lim_{x \rightarrow 0} \frac{1}{x^4 \left(\sqrt{1+\sqrt{1+x^4}} + \sqrt{2} \right) \left(\sqrt{1+x^4} + 1 \right)} \end{aligned}$$

Applying limit $a = \frac{1}{4\sqrt{2}}$

$$\begin{aligned} b &= \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}} \\ &= \lim_{x \rightarrow 0} \frac{(1 - \cos^2 x)(\sqrt{2} + \sqrt{1 + \cos x})}{2 - (1 + \cos x)} \\ b &= \lim_{x \rightarrow 0} (1 + \cos x)(\sqrt{2} + \sqrt{1 + \cos x}) \end{aligned}$$

Applying limits $b = 2(\sqrt{2} + \sqrt{2}) = 4\sqrt{2}$

Now, $ab^3 = \frac{1}{4\sqrt{2}} \times (4\sqrt{2})^3 = 32$

Q4

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{3 + \alpha \sin x + \beta \cos x + \log_e(1-x)}{3 \tan^2 x} &= \frac{1}{3} \\ \Rightarrow \lim_{x \rightarrow 0} \frac{3 + \alpha \left[x - \frac{x^3}{3!} + \dots \right] + \beta \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots \right] + \left(-x - \frac{x^2}{2} - \frac{x^3}{3} \dots \right)}{3 \tan^2 x} &= \frac{1}{3} \\ \Rightarrow \lim_{x \rightarrow 0} \frac{(3 + \beta) + (\alpha - 1)x + \left(-\frac{1}{2} - \frac{\beta}{2} \right) x^2 + \dots}{3x^2} \times \frac{x^2}{\tan^2 x} &= \frac{1}{3} \\ \Rightarrow \beta + 3 = 0, \alpha - 1 = 0 \text{ and } \frac{-\frac{1}{2} - \frac{\beta}{2}}{3} &= \frac{1}{3} \\ \Rightarrow \beta = -3, \alpha = 1 \\ \Rightarrow 2\alpha - \beta = 2 + 3 = 5 \end{aligned}$$

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Solutions

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Q5

Using L'hospital rule

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{0 - \cos x \times 3x^2}{2 \left(x - \frac{\pi}{2} \right)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin \left(x - \frac{\pi}{2} \right)}{2 \left(x - \frac{\pi}{2} \right)} \times \frac{3\pi^2}{4}$$

$$= \frac{3\pi^2}{8}$$

Q6

According to the question ,

$$27r_1 + \frac{9r_2}{2} = -9$$

$$\lim_{x \rightarrow 3} \frac{\int_3^x 8t^2 dt}{\frac{3r_2 x}{2} - r_2 x^2 - r_1 x^3 - 3x}$$

$$= \lim_{x \rightarrow 3} \frac{8x^2}{\frac{3r_2^2}{2} - 2r_2 x - 3r_1 x^2 - 3} \quad (\text{using LH' Rule})$$

$$= \frac{72}{\frac{3r_2}{2} - 6r_2 - 27r_1 - 3}$$

$$= \frac{72}{-\frac{9r_2}{2} - 27r_1 - 3}$$

$$= \frac{72}{9 - 3} = 12$$

Q7

$$\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{\left(\frac{e^{x^2} - 1}{x^2} \right) \times x^2}$$

$$\lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x} \quad \left(\lim_{x \rightarrow 0} \frac{e^{x^2} - 1}{x^2} = 1 \right)$$

$$= \lim_{x \rightarrow 0} \frac{f(x)}{1} \quad (\text{using L Hospital})$$

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Solutions

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$$f(0) = \frac{1}{2}$$

$$\alpha = \frac{1}{2}$$

$$8\alpha^2 = 2$$

Q8

Put $x = 1$

$$\therefore a = 1$$

$$b = \lim_{x \rightarrow 0} \frac{\int_0^x \frac{\ln(1+t)}{1+t^{2024}} dt}{x^2}$$

Using L' HOPITAL Rule

$$b = \lim_{x \rightarrow 0} \frac{\ln(1+x)}{(1+x^{2024})} \times \frac{1}{2x} = \frac{1}{2}$$

$$\text{Now, } cx^2 + dx + e = 0, x^2 + x + 4 = 0$$

(D < 0)

$$\therefore \frac{c}{1} = \frac{d}{1} = \frac{e}{4}$$

Q9

$$\lim_{x \rightarrow 0} \frac{e^{\tan x} - 2|\sin x| - 1}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{e^{4 \sin x} - 2|\sin x| - 1}{|\sin x|^2} \times \frac{\sin^2 x}{x^2}$$

Let $|\sin x| = t$

$$\lim_{t \rightarrow 0} \frac{e^{2t} - 2t - 1}{t^2} \times \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$$

$$= \lim_{t \rightarrow 0} \frac{2e^{2t} - 2}{2t} \times 1 = 2 \times 1 = 2$$

Q10

$$f: R \rightarrow (0, \infty)$$

$$\lim_{x \rightarrow \infty} \frac{f(7x)}{f(x)} = 1$$

 $\therefore f$ is increasing

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Solutions

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$$\begin{aligned} \therefore f(x) &< f(5x) < f(7x) \\ \therefore \frac{f(x)}{f(x)} &< \frac{f(5x)}{f(x)} < \frac{f(7x)}{f(x)} \\ 1 &< \lim_{x \rightarrow \infty} \frac{f(5x)}{f(x)} < 1 \\ \therefore \left[\frac{f(5x)}{f(x)} - 1 \right] & \\ \Rightarrow 1 - 1 &= 0 \end{aligned}$$

Q11

$$\begin{aligned} &ax^2 \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) - b \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right) \\ &+ cx \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) \\ \lim_{x \rightarrow 0} &\frac{x^3 \cdot \frac{\sin x}{x}}{(c-b)x + \left(\frac{b}{2} - c + a \right) x^2 + \left(a - \frac{b}{3} + \frac{c}{2} \right) x^3 + \dots} \\ &= \lim_{x \rightarrow 0} \frac{x^3}{x^3} = 1 \\ c - b &= 0, \quad \frac{b}{2} - c + a = 0 \\ a - \frac{b}{3} + \frac{c}{2} &= 1 \quad a = \frac{3}{4} \quad b = c = \frac{3}{2} \\ a^2 + b^2 + c^2 &= \frac{9}{16} + \frac{9}{4} + \frac{9}{4} \\ 16(a^2 + b^2 + c^2) &= 81 \end{aligned}$$

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